

# Appendix: Analytical Kullback-Leibler Divergence between von Mises and Wrapped Cauchy Distributions

## 1 Introduction

This appendix presents the analytical derivations of the Kullback-Leibler (KL) divergences between the von Mises distribution  $P$  and the wrapped Cauchy distribution  $Q$  on the circle  $S^1$ . The densities are defined as:

$$p(\theta) = \frac{1}{2\pi I_0(\kappa)} e^{\kappa \cos(\theta - \mu_P)}, \quad (1)$$

$$q(\theta) = \frac{1 - \rho^2}{2\pi(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))}, \quad (2)$$

where  $\mu_P, \mu_Q \in [-\pi, \pi)$  are the mean directions,  $\kappa > 0$  and  $\rho \in [0, 1)$  are the concentration parameters, and  $I_n(\kappa)$  is the modified Bessel function of the first kind of order  $n$ .

## 2 Key Mathematical Preparations

### 2.1 Trigonometric Moments

Using standard integrations, the trigonometric moments for  $P$  and  $Q$  with respect to their centered directions are given by:

$$E_P[\cos(n(\theta - \mu_P))] = \frac{I_n(\kappa)}{I_0(\kappa)}, \quad E_P[\sin(n(\theta - \mu_P))] = 0, \quad (3)$$

$$E_Q[\cos(n(\theta - \mu_Q))] = \rho^n, \quad E_Q[\sin(n(\theta - \mu_Q))] = 0, \quad (4)$$

for any integer  $n \geq 1$ .

### 2.2 Fourier Series Expansion of the Log-Denominator

To expand the log-denominator of  $q(\theta)$ , let  $z = \rho e^{ix}$  be a complex number with  $|z| = \rho < 1$ . Using the Taylor series expansion of the complex logarithm

$\log(1 - z) = -\sum_{n=1}^{\infty} \frac{z^n}{n}$ , we have:

$$\log(1 - \rho e^{ix}) = -\sum_{n=1}^{\infty} \frac{\rho^n}{n} e^{inx}. \quad (5)$$

Adding this to its complex conjugate yields:

$$\log(1 - \rho e^{ix}) + \log(1 - \rho e^{-ix}) = -2 \sum_{n=1}^{\infty} \frac{\rho^n}{n} \cos(nx). \quad (6)$$

Since the left-hand side can be combined using logarithmic properties as:

$$\log((1 - \rho e^{ix})(1 - \rho e^{-ix})) = \log(1 + \rho^2 - 2\rho \cos x), \quad (7)$$

we obtain the desired Fourier series representation:

$$\log(1 + \rho^2 - 2\rho \cos x) = -2 \sum_{n=1}^{\infty} \frac{\rho^n}{n} \cos(nx). \quad (8)$$

### 3 Derivation of $D_{\text{KL}}(P \parallel Q)$

The KL divergence from  $P$  to  $Q$  is defined as  $D_{\text{KL}}(P \parallel Q) = E_P[\log p(\theta)] - E_P[\log q(\theta)]$ . First, the expectation of the log-density of  $P$  is:

$$E_P[\log p(\theta)] = -\log(2\pi) - \log I_0(\kappa) + \kappa \frac{I_1(\kappa)}{I_0(\kappa)}. \quad (9)$$

Next, for the log-density of  $Q$ , we apply the Fourier expansion of the denominator with  $x = \theta - \mu_Q$ . By the trigonometric addition formula and the prepared moments under  $P$ , we obtain:

$$E_P[\cos(n(\theta - \mu_Q))] = \frac{I_n(\kappa)}{I_0(\kappa)} \cos(n(\mu_P - \mu_Q)). \quad (10)$$

Therefore, we have:

$$E_P[\log q(\theta)] = \log(1 - \rho^2) - \log(2\pi) + 2 \sum_{n=1}^{\infty} \frac{\rho^n I_n(\kappa)}{n I_0(\kappa)} \cos(n(\mu_P - \mu_Q)). \quad (11)$$

Combining these terms yields:

$$D_{\text{KL}}(P \parallel Q) = \kappa \frac{I_1(\kappa)}{I_0(\kappa)} - \log I_0(\kappa) - \log(1 - \rho^2) - 2 \sum_{n=1}^{\infty} \frac{\rho^n I_n(\kappa)}{n I_0(\kappa)} \cos(n(\mu_P - \mu_Q)). \quad (12)$$

## 4 Derivation of $D_{\text{KL}}(Q \parallel P)$

The KL divergence from  $Q$  to  $P$  is  $D_{\text{KL}}(Q \parallel P) = E_Q[\log q(\theta)] - E_Q[\log p(\theta)]$ . Under  $Q$ , using  $E_Q[\cos(n(\theta - \mu_Q))] = \rho^n$ , we evaluate the expectation of the log-denominator of  $Q$ :

$$E_Q[\log(1 + \rho^2 - 2\rho \cos(\theta - \mu_Q))] = -2 \sum_{n=1}^{\infty} \frac{\rho^{2n}}{n} = 2 \log(1 - \rho^2). \quad (13)$$

This simplifies the self-entropy term to:

$$E_Q[\log q(\theta)] = -\log(1 - \rho^2) - \log(2\pi). \quad (14)$$

Next, for the expectation of the log-density of  $P$ , by the trigonometric addition formula and the prepared moments under  $Q$ , we obtain:

$$E_Q[\cos(\theta - \mu_P)] = \rho \cos(\mu_P - \mu_Q). \quad (15)$$

Hence, the cross-entropy term is:

$$E_Q[\log p(\theta)] = -\log(2\pi) - \log I_0(\kappa) + \kappa \rho \cos(\mu_P - \mu_Q). \quad (16)$$

Combining these results yields:

$$D_{\text{KL}}(Q \parallel P) = \log I_0(\kappa) - \log(1 - \rho^2) - \kappa \rho \cos(\mu_P - \mu_Q). \quad (17)$$